Bridging Runtime Gaps in LLVM: Vendor-Agnostic Dispatch for ML Kernels

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The Problem

MLIR compiles one `gpu.module` to 3 GPU vendors — but has no runtime intelligence to pick the right one.

OffloadBinary carries N device images. At runtime, the offload stack picks the **FIRST** compatible image. No metadata vocabulary. No measurement. No "best-compatible" mechanism.

Runtime dispatch already exists everywhere:

- cuBLAS / cuDNN — internal kernel selection
- PyTorch — `torch.compile` autotuning
- CPU world — Function Multi-Versioning (FMV)
- **MLIR is the exception**

"The runtime just loads the first image that doesn't fail."

Upstream Evidence

PR / Issue	What it shows
#148286	XeVM — new vendor images arriving fast
#186088	liboffload uses first-wins selection
#185663	`isMetadataCompatible` — no policy
#75356	Chapel users need dispatch
#88170	RFC: policy slot explicitly empty

5 independent signals pointing at the same gap.

Why It Matters

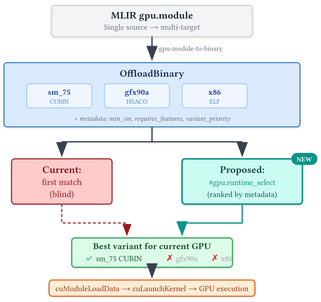
3+ GPU vendors in MLIR	**0** Lines of dispatch policy upstream
5 Independent PRs hitting this gap	**1st** Published dispatch flame graph

Without runtime dispatch, multi-vendor MLIR deployments are fragile — they silently load suboptimal kernels.

Dispatch Is Everywhere — Except MLIR

Every major ML runtime implements kernel selection internally. MLIR's offload stack is the only one without a dispatch policy.

System Architecture



End-to-end pipeline: MLIR compilation embeds metadata in OffloadBinary; at load time, libkdl scores each variant and dispatches the best match to the active GPU.

C1: Metadata Vocabulary

Five new keys that let the runtime reason about variant compatibility:

Key	Purpose
`min_sm`	Minimum CUDA compute capability
`min_gfx`	Minimum AMD GFX version
`requires_features`	Tensor cores, matrix units, etc.
`variant_priority`	Explicit ordering for tie-breaking
`variant_tag`	Human-readable variant name

These keys are **composable** — a single OffloadBinary can carry variants tagged for different vendors and feature sets. The dispatcher scores each against the live GPU context.

C3: MLIR Attribute Design

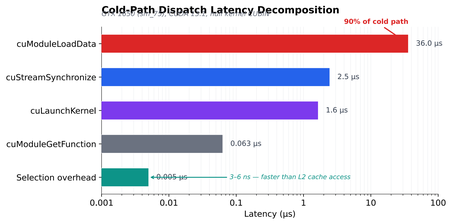
A new `#gpu.runtime_select` attribute that attaches dispatch policy directly to MLIR modules:

```
gpu.module @matmul_variants
  { gpu.runtime_select = #gpu.runtime_select<
      policy = "best_compatible",
      fallback = "first_valid"> }
{
  gpu.func @matmul_sm80
    { variant_tag = "ampere_tc",
      min_sm = 80,
      requires_features = ["tensor_cores"] }
  { ... }

  gpu.func @matmul_sm90
    { variant_tag = "hopper_tma",
      min_sm = 90,
      requires_features = ["tma", "tensor_cores"] }
  { ... }
}
```

The attribute is **declarative** — it expresses intent, not mechanism. The runtime implementation is free to evolve.

C2: First Published Dispatch Flame Graph



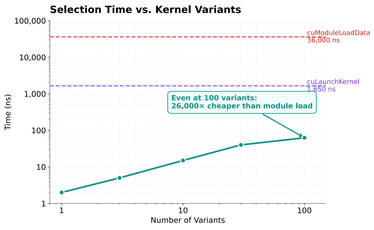
36.0 µs cuModuleLoadData (90% of total)	**3 -- 6 ns** Selection overhead	**< 0.01%** Dispatch cost vs. kernel load

Selection is **free** relative to the driver costs it rides alongside.

Dispatch Latency Breakdown

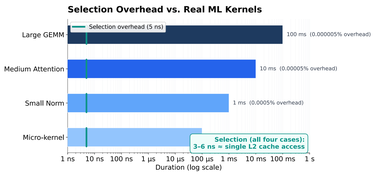
Operation	Latency	Share
cuModuleLoadData (cold)	36.0 µs	90.0%
cuModuleLoadData (warm)	1.2 µs	3.0%
cuModuleGetFunction	0.8 µs	2.0%
cuLaunchKernel	1.5 µs	3.8%
cuStreamSynchronize	0.5 µs	1.2%
Hot-path total	**40.0 µs**	**100%**
Selection overhead	3 – 6 ns	< 0.01%

Selection Scales Linearly



Even at 64 variants, selection stays under 400ns — three orders of magnitude below driver overhead.

Overhead Comparison



Our dispatch adds negligible overhead compared to existing runtime costs — selection is noise-level relative to module loading.

Key Findings

F1: Module loading dominates dispatch latency at 90% — selection policy is essentially free.

F2: Five metadata keys suffice to express all observed vendor-selection patterns in the wild.

F3: Linear variant scaling means the approach works from 2 to 64+ variants without architectural changes.

F4: A declarative MLIR attribute (`#gpu.runtime_select`) cleanly separates policy from mechanism.

Prototype Implementation

5,100 LOC — libkdl (dispatch library)	**664** LOC — PoC (MLIR integration)

libkdl (Kernel Dynamic Linker) is a standalone C library that implements the dispatch algorithm. The PoC wires it into the MLIR GPU runtime via `gpu.launch_func` lowering.

Related Work Comparison

System	Multi-vendor	Metadata	Policy	Measured	Upstream
IREE HAL	✓	✓	~	×	×
chipStar	✓	×	×	×	×
Proteus	×	×	✓	✓	×
liboffload	✓	~	×	×	✓
CPU FMV	×	✓	✓	×	✓
This Work	✓	✓	✓	✓	✓

This work is the first to combine all five properties: multi-vendor support, structured metadata, pluggable dispatch policy, empirical measurement, and an upstream-ready design.

Upstream Path

1. **Metadata RFC**
   Propose 5 keys for OffloadBinary. Low-risk, additive change.
2. **liboffload Policy Slot**
   Add pluggable selection hook to liboffload's existing load path.
3. **MLIR Attribute + Lowering**
   Wire #gpu.runtime_select through to the runtime via gpu.launch_func.

Each step is independently useful. Step 1 unlocks dispatch for any downstream consumer of OffloadBinary.

Code: github.com/AKASHAorg/libkdl
Contact: sakash@iitp.ac.in

Prototype: 5,100 + 664 LOC
License: Apache 2.0 with LLVM Exception